

Embedded System Certification Course for GosNIIAS

Assignment 1

marinvent CORPORATION

Issued: **December 22, 2011**

Due Date: **N/A**

Conduct: **Individual**

Requirements: **Internet access and a web browser**

This assignment applies the concepts learned from the System Certification and RTCA/DO-160G lectures to the Integrated Modular Architecture (IMA) Avionics system, which will be revisited throughout the course. Subsequent lectures, case studies, and labs, will build on the research you perform for this assignment, so try to answer the questions concisely but comprehensively.

Q1. (Based on 1.4 – System Certification)

The IMA will be certified by Transport Canada (TC) and the Federal Aviation Administration (FAA The IMA chassis will eventually house avionics cards that will perform a variety of system functions, including Global Positioning (GPS), Flight Management (FMS), and Fuel Flow (FF).

1) Identify all TSOs and Aeronautical Circulars (ACs) applicable to the following sub-functions included in the case-study IMA:
   a) GPS
   b) FMS
   c) Fuel Flow
2) Based on the criticality definitions in DO-178B and ARP 4754, what criticality might each of these sub-systems have? Briefly justify your responses.

Q2. (Based on 2.1 – RTCA/DO-160G).

An Ng engine tachometer (RPM indicator) is fitted on the instrument panel as part of a P&WC PT6A engine installation on an executive turboprop aircraft. This RPM indicating system is qualified under RTCA/DO-160D for vibration as: **[SBM]**. The manufacturer proposes to use the same qualified system on a PT6A engine installation in an executive turbine helicopter, and you are the delegate responsible for the helicopter's powerplant certification.

1) What are your comments on the applicant's plan?
2) Complete the environmental qualification form (Figure A-1[1]) for the target helicopter installation. Include a short justification for each of your entries.

[1] Source: RTCA/DO-160G

Figure A-1 Environmental Qualification Form

NOMENCLATURE: _____
TYPE/MODEL/PART NO: _____ TSO NUMBER _____

MANUFACTURER'S SPECIFICATION AND/OR OTHER APPLICABLE SPECIFICATION: _____
_____

MANUFACTURER: _____
ADDRESS: _____
_____

REVISION & CHANGE NUMBER OF DO-160: _____ **DATE TESTED:** _____

CONDITIONS 1/	SECTION	DESCRIPTION OF TESTS CONDUCTED
Temperature and Altitude	4.0	
Low Temperature	4.5.1	
High Temperature	4.5.2 & 4.5.3	
In-Flight Loss of Cooling	4.5.4	
Altitude	4.6.1	
Decompression	4.6.2	
Overpressure	4.6.3	
Temperature Variation	5.0	
Humidity	6.0	
Operational Shock and Crash Safety	7.0	
Vibration	8.0	
Explosion	9.0	
Waterproofness	10.0	
Fluids Susceptibility	11.0	
Sand and Dust	12.0	
Fungus	13.0	
Salt Spray	14.0	
Magnetic Effect	15.0	
Power Input	16.0	
Voltage Spike	17.0	
Audio Frequency Susceptibility	18.0	
Induced Signal Susceptibility	19.0	
Radio Frequency Susceptibility	20.0	

1/ The information listed below provides examples only. It is not intended to be a comprehensive listing of all test conditions.

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Appendix A
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FIGURE A-1 CONCLUDED

CONDITIONS	SECTION	DESCRIPTION OF TESTS CONDUCTED
Radio Frequency Emission	21.0	
Lightning Induced Transient Susceptibility	22.0	
Lightning Direct Effects	23.0	
Icing	24.0	
Electrostatic Discharge	25.0	
Other Tests		

REMARKS

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- Special Conditions: Include power, special cooling, installation instructions, etc.

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– End of Assignment –